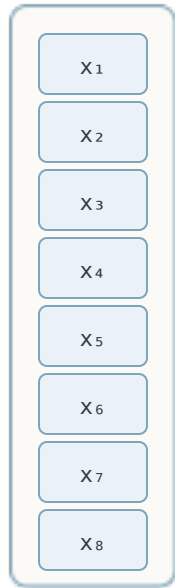


SOM training: best-matching unit and Gaussian neighborhood

Each input vector selects a winning neuron; nearby codebook vectors are pulled toward the input

INPUT VECTOR

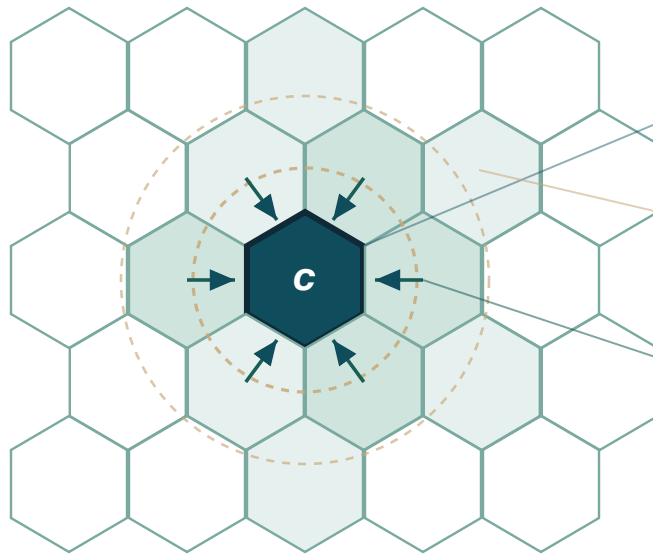


X

8 temporal features

present x →

5 × 5 HEXAGONAL NEURON GRID



c
BMU

Best-matching unit

$$c = \operatorname{argmin}_j \|x - w_j\|$$

$h_{cj}(t)$

Gaussian neighborhood

influence decays with grid distance

w_j

Codebook weights

pulled toward x by $\alpha h(x - w)$

$$w_j(t+1) = w_j(t) + \alpha(t) \cdot h_{cj}(t) \cdot (x - w_j)$$

Neurons nearer the BMU c are adapted more strongly; the pull fades outward.